

MATH 221 — Exam 1

1. calculus/derivatives/chain-rule-01 *Chain rule with trig*

date	course	term	type	assignment
2026-09-21	MATH 221	Fall 2026	worksheet	Worksheet 3

Differentiate each function.

(a) (4 points) $f(x) = \sin(x^2)$

(b) (6 points) $g(x) = \cos^3(2x)$

2. (10 points) calculus/integrals/u-sub-01 *Substitution*

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Evaluate $\int x\sqrt{x^2+1} \, dx$.

3. (6 points) calculus/limits/squeeze-01 *Squeeze theorem*

Not used before.

Use the squeeze theorem to evaluate $\lim_{x \rightarrow 0} x^2 \cos(1/x)$.